

SOLUTION

The problem can be solved using **recursive predicates** in Prolog.

- If $N = 0$, the sum is 0.
- For $N > 0$, calculate the sum up to $N-1$ and add N to it.
- Continue until the base case $N = 0$ is reached.

For example:

$$\text{sum}(5) = 1 + 2 + 3 + 4 + 5 = 15$$